
LOCAL CREDIT, GLOBAL SETTLEMENT: ENHANCING TRUST-BASED CREDIT CREATION WITH ON-CHAIN RESERVE CONTRACTS

Abstract

In this work, we propose Basis, a framework where both local trust-based (and IOU-based) currencies as well as blockchain-based trust-minimized global ones can coexist via smart contract powered reserves needed when there is no trust (possibly connecting local trading circles). Coordination with blockchain is needed only during debt note redemption against a reserve contract. Payments (debt note transfers) are done offchain, so with low fees and no need to use low-security centralized blockchain-like systems with high throughput. For coordination and transparency of debt in the system, a tracker service is used, which can not steal money from reserves. We provide analysis of trust-minimization in regards with tracker activities.

Current offchain payment systems backed by on-chain reserves (Lightning Network, Cashu, Fedimint) require full backing and so do not allow for credit creation. In contrast, Basis allows for credit creation with no requirements set for reserves at the protocol level. Then it can be used for community trading without using blockchain (and possibly, without Internet even, just by using a local mesh network), but when trust is insufficient to expand credit, on-chain reserves can be established and used. We expect that on a local scale trust-based economic relationships would dominate, and on a global scale (e.g. when one community is doing trades with another) full coverage with reserves would be needed in most cases. The whole system will use limited in disconnected from real-world needs supply blockchain assets only where they are needed (there is no trust), while enabling monetary expansion whenever possible (so where peer-to-peer trust can be established), thus allowing to create a viable alternative to political money.

The main payment unit is IOU note, which is signed by the issuer and the tracker. Our design allows for debt transferability, if issuer agrees on that, so if peer A issued debt to peer B, and peer B pays with it, fully or partially, to peer C, with A also co-signing, and then A owes C. We show that this design allows for minimal trust to tracker service. A tracker service is committing its state on the blockchain periodically. If a tracker service ceases to exist, it is possible to redeem debt notes against on-chain reserves

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using this committed state. There could be multiple trackers around the world. We consider different designs, from just centralized server, to federated control, to rollups and sidechains.

We provide implementation of reserve contract as well as offchain clients (tracker server and example clients). We show an example of group trading over mesh network with occasional Internet connection. Another example shows AI agent economies where autonomous agents create credit relationships for services.

Keywords: blockchain, digital currency, peer-to-peer money, trust-based credit, IOU notes

1 Introduction

Peer-to-peer monetary systems, such as ChainCash [1], enable decentralized money creation through trust and collateral. While on-chain protocols provide transparency and censorship resistance, they often struggle with scalability and cost when every transaction must be recorded on a public blockchain. Off-chain alternatives, such as payment channels and Layer 2 solutions, improve scalability but introduce new trust assumptions and state management challenges.

In this work, we propose Basis, a trust-minimized state tracker designed to complement peer-to-peer monetary systems. Basis maintains an off-chain AVL tree of debt relationships between economic agents, witnesses new notes by signing them, and periodically commits state digests to a public blockchain. This design allows for efficient note circulation off-chain while preserving the security properties of on-chain redemption.

The paper is organized as follows. In Section 2 we describe the design of Basis. Section 3 analyzes security properties. Section 4 provides implementation details. In Section 5 we provide analysis of advantages and drawbacks.

2 Design

We consider a set of economic agents $a_1, \dots, a_n$ participating in a peer-to-peer monetary system. Each agent may issue IOU notes, creating debt relationships with other agents. Basis tracks these relationships in an AVL tree mapping:

$$\text{Blake2b256}(\text{payerKey} || \text{payeeKey}) \rightarrow (\text{totalDebt}, \text{timestamp})$$

2.1 Note Witnessing

When an agent issues a new IOU note, the note contains the issuer’s signature and the intended recipient. Before the recipient accepts the note, it may request witnessing from a Basis tracker.

The tracker verifies the note, updates its local state to include the new debt relationship, and returns a signature certifying that the debt is now tracked.

2.2 State Commitment

Periodically, the tracker commits the root digest of its AVL tree to the public blockchain. This commitment serves two purposes: it provides a timestamped checkpoint of the global debt state, and it enables emergency redemption in case the tracker becomes unavailable.

2.3 Debt Transfer

When an agent wishes to transfer a portion of its debt to a new creditor, the original debtor signs new notes, the tracker witnesses both the reduction in the old debt and the creation of the new debt, and the old note is cancelled. This enables triangular trade and debt restructuring without requiring full redemption.

3 Security Analysis

3.1 Tracker Cannot Steal Funds

The tracker only certifies inclusion of notes in its state. It cannot unilaterally create or redeem notes. The issuer's signature is always required for redemption, ensuring that the tracker cannot steal funds.

3.2 Emergency Exit

If the tracker becomes unavailable, an emergency exit mechanism activates after a timeout period (e.g., 3 days or 2160 blocks). After this period, the tracker signature becomes optional for redemption. The same message format is used: `key||totalDebt||timestamp`. Replay attacks are prevented by timestamp verification, and the reserve owner's signature is still required.

3.3 Anti-Censorship

Previously witnessed notes can still be redeemed even if the tracker refuses to witness new notes. The last committed state on the blockchain provides a floor for redemption rights.

4 Implementation

We provide implementation of the Basis protocol in software, including smart contracts for on-chain reserves, a tracker server for coordinating off-chain state, and client libraries for issuing and managing IOU notes.

4.1 Reserve Contract

The reserve contract is implemented on the Ergo blockchain using the ErgoScript smart contract language. It locks collateral (ERG or custom tokens) and allows redemption when presented with a valid debt note and signatures. The contract enforces the timeout rules for emergency redemption without tracker signatures.

4.2 Tracker Server

The tracker server maintains the AVL tree of debt relationships, witnesses new notes by signing them, and periodically commits state digests to the blockchain. It exposes APIs for note witnessing, state queries, and commitment retrieval. The server publishes alerts via the NOSTR protocol for collateralization monitoring.

4.3 Client Libraries

Client libraries provide functionality for agents to issue IOU notes, request tracker witnessing, transfer debt to new creditors, and redeem notes against reserves. The libraries handle signature generation, note serialization, and communication with tracker and blockchain nodes.

4.4 Examples

We demonstrate two example applications: group trading over a local mesh network with occasional Internet connection, where notes circulate off-chain and synchronize when connectivity is available; and AI agent economies where autonomous agents create credit relationships for services rendered.

5 Advantages and Drawbacks

- Scalable off-chain tracking with on-chain security guarantees
- Emergency exit prevents lock-in to a single tracker

- Anti-censorship protections for previously witnessed notes
- Periodic commitments enable global state verification

References

- [1] Alexander Chepurnoy. Chaincash: Money creation with elastic supply via trust and blockchain assets. <https://github.com/ChainCashLabs/chaincash>. Accessed: 2024-01-01.